



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

Re - accredited by NAAC with 'A' Grade

Founder: Prof. Dr. S. B. Mujumdar, M.Sc., Ph.D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name : Mathematics-II
Course Code : TE7681
Faculty : Engineering
Course Credit : 4
Course Level : 2
Sub-Committee (Specialization) : Applied Sciences
Learning Objectives :

2. Define the convergence of series of positive terms, alternating series and power series by using appropriate tests, to expand periodic functions as Fourier series..

3. Identify and solve linear differential equations using suitable method and apply to solve engineering problems

The students will be able to:

1. Explain double and triple integrals in cartesian form and evaluate gradient, direction derivative, curl and divergence.

4. Apply the basic knowledge of complex analysis in complex differentiation and integration.

5. Find the bilinear transformations and discuss the conformal mapping.

Books Recommended :

Book	Author	Publisher
Advanced Engineering Mathematics	R. K. Jain and S. R. K. Iyengar	Narosa; 1st edition (22 May 2002)
Advanced Engineering Mathematics	Erwin Kreyszig	Wiley; 10th edition (1 January 2015)
Calculus and Analytical geometry	G.B. Thomas and R.L. Finney	Pearson Education India (30 January 2010)
Engineering Mathematics	N. P. Bali and Manish Goyal	Laxmi Publications; Ninth edition (1 January 2016)
Higher Engineering Mathematics	B. S. Grewal	KHANNA PUBLISHERS; Forty Fourth edition (1 January 1965)

Course Outline :

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Multiple Integration and Vector differentiation: Double and triple integrals with cartesian coordinates, Gradient, Directional derivative, Curl and Divergence.	9	9
2	Sequence and Series: Sequences, Convergence of sequences, Series, Tests of convergence: P - series, Comparison test, D'Alembert's ratio test, Raabe's test, Cauchy's nth root test, Alternating series, Absolute and conditional convergence, Power series, Fourier series, Half range Fourier series.	10	10
3	Ordinary Differential Equations:	10	10

	Exact differential equations, Linear differential equations with constant coefficients, Method of variation of parameters, Cauchy and Legendre's linear differential equations, Simultaneous differential equations. Applications: Electrical circuits, Rectilinear motion.		
4	Functions of a Complex Variable: Definition of a function of a complex variable, Elementary functions, Limits and continuity, Differentiability, Analyticity, Cauchy-Riemann equations, Laplace equation, Harmonic functions, Line integrals, Contour integration, Cauchy's Integral theorem, Cauchy's integral formulae residues, Cauchy's residue theorem.	10	10
5	Conformal Mapping: Transformations of exponential, trigonometric, inverse and logarithmic functions, Bilinear and inverse transformations.	6	6
6	Tutorial Tutorial will be conducted on all topics.	15	15
Total		60	60

Pre Requisites :

Mathematics-I

Evaluation :

- A. Continuous Assessment
 - 1. Unit Tests
 - 2. Tutorials
 - 3. Class test
 - 4. Assignments
- B. End Semester Examination
 - 1. Written Examination

Pedagogy :

- 1. Interactive sessions
- 2. Class room teaching
- 3. Tutorials
- 4. Assignment